Windows Azure Storage: A Highly Available Cloud Storage Service with Strong Consistency

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Abstract

Windows Azure Storage (WAS) is a cloud storage system that provides customers the ability to store seemingly limitless amounts of data for any duration of time. WAS customers have access to their data from anywhere at any time and only pay for what they use and store. In WAS, data is stored durably using both local and geographic replication to facilitate disaster recovery. Currently, WAS storage comes in the form of Blobs (files), Tables (structured storage), and Queues (message delivery). In this paper, we describe the WAS architecture, global namespace, and data model, as well as its resource provisioning, load balancing, and replication systems.

Categories and Subject Descriptors

D.4.2 [**Operating Systems**]: Storage Management—*Secondary storage*; D.4.3 [**Operating Systems**]: File Systems Management—*Distributed file systems*; D.4.5 [**Operating Systems**]: Reliability—*Fault tolerance*; D.4.7 [**Operating Systems**]: Organization and Design—*Distributed systems*; D.4.8 [**Operating Systems**]: Performance—*Measurements*

General Terms

Algorithms, Design, Management, Measurement, Performance, Reliability.

Keywords

Cloud storage, distributed storage systems, Windows Azure.

1. Introduction

Windows Azure Storage (WAS) is a scalable cloud storage system that has been in production since November 2008. It is used inside Microsoft for applications such as social networking search, serving video, music and game content, managing medical records, and more. In addition, there are thousands of customers outside Microsoft using WAS, and anyone can sign up over the Internet to use the system.

WAS provides cloud storage in the form of Blobs (user files), Tables (structured storage), and Queues (message delivery). These three data abstractions provide the overall storage and workflow for many applications. A common usage pattern we see is incoming and outgoing data being shipped via Blobs, Queues providing the overall workflow for processing the Blobs, and intermediate service state and final results being kept in Tables or Blobs.

An example of this pattern is an ingestion engine service built on Windows Azure to provide near real-time Facebook and Twitter search. This service is one part of a larger data processing pipeline that provides publically searchable content (via our search engine, Bing) within 15 seconds of a Facebook or Twitter user's posting or status update. Facebook and Twitter send the raw public content to WAS (e.g., user postings, user status updates, etc.) to be made publically searchable. This content is stored in WAS Blobs. The ingestion engine annotates this data with user auth, spam, and adult scores; content classification; and classification for language and named entities. In addition, the engine crawls and expands the links in the data. While processing, the ingestion engine accesses WAS Tables at high rates and stores the results back into Blobs. These Blobs are then folded into the Bing search engine to make the content publically searchable. The ingestion engine uses Queues to manage the flow of work, the indexing jobs, and the timing of folding the results into the search engine. As of this writing, the ingestion engine for Facebook and Twitter keeps around 350TB of data in WAS (before replication). In terms of transactions, the ingestion engine has a peak traffic load of around 40,000 transactions per second and does between two to three billion transactions per day (see Section 7 for discussion of additional workload profiles).

In the process of building WAS, feedback from potential internal and external customers drove many design decisions. Some key design features resulting from this feedback include:

Strong Consistency – Many customers want strong consistency: especially enterprise customers moving their line of business applications to the cloud. They also want the ability to perform conditional reads, writes, and deletes for optimistic concurrency control [12] on the strongly consistent data. For this, WAS provides three properties that the CAP theorem [2] claims are difficult to achieve at the same time: strong consistency, high availability, and partition tolerance (see Section 8).

Global and Scalable Namespace/Storage – For ease of use, WAS implements a global namespace that allows data to be stored and accessed in a consistent manner from any location in the world. Since a major goal of WAS is to enable storage of massive amounts of data, this global namespace must be able to address exabytes of data and beyond. We discuss our global namespace design in detail in Section 2.

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Disaster Recovery – WAS stores customer data across multiple data centers hundreds of miles apart from each other. This redundancy provides essential data recovery protection against disasters such as earthquakes, wild fires, tornados, nuclear reactor meltdown, etc.

Multi-tenancy and Cost of Storage – To reduce storage cost, many customers are served from the same shared storage infrastructure. WAS combines the workloads of many different customers with varying resource needs together so that significantly less storage needs to be provisioned at any one point in time than if those services were run on their own dedicated hardware.

We describe these design features in more detail in the following sections. The remainder of this paper is organized as follows. Section 2 describes the global namespace used to access the WAS Blob, Table, and Queue data abstractions. Section 3 provides a high level overview of the WAS architecture and its three layers: Stream, Partition, and Front-End layers. Section 4 describes the stream layer, and Section 5 describes the partition layer. Section 6 shows the throughput experienced by Windows Azure applications accessing Blobs and Tables. Section 7 describes some internal Microsoft workloads using WAS. Section 8 discusses design choices and lessons learned. Section 9 presents related work, and Section 10 summarizes the paper.

2. Global Partitioned Namespace

A key goal of our storage system is to provide a single global namespace that allows clients to address all of their storage in the cloud and scale to arbitrary amounts of storage needed over time. To provide this capability we leverage DNS as part of the storage namespace and break the storage namespace into three parts: an account name, a partition name, and an object name. As a result, all data is accessible via a URI of the form:

http(s)://**AccountName**.<service>[1].core.windows.net/**PartitionName**/**ObjectName**

The **AccountName** is the customer selected account name for accessing storage and is part of the DNS host name. The AccountName DNS translation is used to locate the primary storage cluster and data center where the data is stored. This primary location is where all requests go to reach the data for that account. An application may use multiple AccountNames to store its data across different locations.

In conjunction with the AccountName, the **PartitionName** locates the data once a request reaches the storage cluster. The PartitionName is used to scale out access to the data across storage nodes based on traffic needs.

When a PartitionName holds many objects, the **ObjectName** identifies individual objects within that partition. The system supports atomic transactions across objects with the same PartitionName value. The ObjectName is optional since, for some types of data, the PartitionName uniquely identifies the object within the account.

This naming approach enables WAS to flexibly support its three data abstractions[2]. For Blobs, the full blob name is the PartitionName. For Tables, each entity (row) in the table has a primary key that consists of two properties: the PartitionName and the ObjectName. This distinction allows applications using Tables to group rows into the same partition to perform atomic transactions across them. For Queues, the queue name is the PartitionName and each message has an ObjectName to uniquely identify it within the queue.

3. High Level Architecture

Here we present a high level discussion of the WAS architecture and how it fits into the Windows Azure Cloud Platform.

3.1 Windows Azure Cloud Platform

The Windows Azure Cloud platform runs many cloud services across different data centers and different geographic regions. The Windows Azure Fabric Controller is a resource provisioning and management layer that provides resource allocation, deployment/upgrade, and management for cloud services on the Windows Azure platform. WAS is one such service running on top of the Fabric Controller.

The Fabric Controller provides node management, network configuration, health monitoring, starting/stopping of service instances, and service deployment for the WAS system. In addition, WAS retrieves network topology information, physical layout of the clusters, and hardware configuration of the storage nodes from the Fabric Controller. WAS is responsible for managing the replication and data placement across the disks and load balancing the data and application traffic within the storage cluster.

3.2 WAS Architectural Components

An important feature of WAS is the ability to store and provide access to an immense amount of storage (exabytes and beyond). We currently have 70 petabytes of raw storage in production and are in the process of provisioning a few hundred more petabytes of raw storage based on customer demand for 2012.

The WAS production system consists of Storage Stamps and the Location Service (shown in Figure 1).

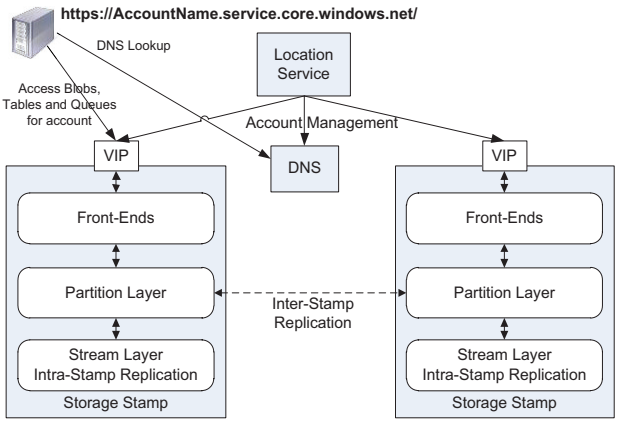

Figure 1: High-level architecture

Storage Stamps – A storage stamp is a cluster of *N* racks of storage nodes, where each rack is built out as a separate fault domain with redundant networking and power. Clusters typically range from 10 to 20 racks with 18 disk-heavy storage nodes per rack. Our first generation storage stamps hold approximately 2PB of raw storage each. Our next generation stamps hold up to 30PB of raw storage each.

[1] <service> specifies the service type, which can be **blob**, **table**, or **queue**.

[2] APIs for Windows Azure Blobs, Tables, and Queues can be found here: http://msdn.microsoft.com/en-us/library/dd179355.aspx

To provide low cost cloud storage, we need to keep the storage provisioned in production as highly utilized as possible. Our goal is to keep a storage stamp around 70% utilized in terms of capacity, transactions, and bandwidth. We try to avoid going above 80% because we want to keep 20% in reserve for (a) disk short stroking to gain better seek time and higher throughput by utilizing the outer tracks of the disks and (b) to continue providing storage capacity and availability in the presence of a rack failure within a stamp. When a storage stamp reaches 70% utilization, the location service migrates accounts to different stamps using inter-stamp replication (see Section 3.4).

Location Service (LS) – The location service manages all the storage stamps. It is also responsible for managing the account namespace across all stamps. The LS allocates accounts to storage stamps and manages them across the storage stamps for disaster recovery and load balancing. The location service itself is distributed across two geographic locations for its own disaster recovery.

WAS provides storage from multiple locations in each of the three geographic regions: North America, Europe, and Asia. Each location is a data center with one or more buildings in that location, and each location holds multiple storage stamps. To provision additional capacity, the LS has the ability to easily add new regions, new locations to a region, or new stamps to a location. Therefore, to increase the amount of storage, we deploy one or more storage stamps in the desired location's data center and add them to the LS. The LS can then allocate new storage accounts to those new stamps for customers as well as load balance (migrate) existing storage accounts from older stamps to the new stamps.

Figure 1 shows the location service with two storage stamps and the layers within the storage stamps. The LS tracks the resources used by each storage stamp in production across all locations. When an application requests a new account for storing data, it specifies the location affinity for the storage (e.g., US North). The LS then chooses a storage stamp within that location as the primary stamp for the account using heuristics based on the load information across all stamps (which considers the fullness of the stamps and other metrics such as network and transaction utilization). The LS then stores the account metadata information in the chosen storage stamp, which tells the stamp to start taking traffic for the assigned account. The LS then updates DNS to allow requests to now route from the name https://AccountName.service.core.windows.net/ to that storage stamp's virtual IP (VIP, an IP address the storage stamp exposes for external traffic).

3.3 Three Layers within a Storage Stamp

Also shown in Figure 1 are the three layers within a storage stamp. From bottom up these are:

Stream Layer – This layer stores the bits on disk and is in charge of distributing and replicating the data across many servers to keep data durable within a storage stamp. The stream layer can be thought of as a distributed file system layer within a stamp. It understands files, called "streams" (which are ordered lists of large storage chunks called "extents"), how to store them, how to replicate them, etc., but it does not understand higher level object constructs or their semantics. The data is stored in the stream layer, but it is accessible from the partition layer. In fact, partition servers (daemon processes in the partition layer) and stream servers are co-located on each storage node in a stamp.

Partition Layer – The partition layer is built for (a) managing and understanding higher level data abstractions (Blob, Table, Queue), (b) providing a scalable object namespace, (c) providing transaction ordering and strong consistency for objects, (d) storing object data on top of the stream layer, and (e) caching object data to reduce disk I/O.

Another responsibility of this layer is to achieve scalability by partitioning all of the data objects within a stamp. As described earlier, all objects have a PartitionName; they are broken down into disjointed ranges based on the PartitionName values and served by different partition servers. This layer manages which partition server is serving what PartitionName ranges for Blobs, Tables, and Queues. In addition, it provides automatic load balancing of PartitionNames across the partition servers to meet the traffic needs of the objects.

Front-End (FE) layer – The Front-End (FE) layer consists of a set of stateless servers that take incoming requests. Upon receiving a request, an FE looks up the AccountName, authenticates and authorizes the request, then routes the request to a partition server in the partition layer (based on the PartitionName). The system maintains a Partition Map that keeps track of the PartitionName ranges and which partition server is serving which PartitionNames. The FE servers cache the Partition Map and use it to determine which partition server to forward each request to. The FE servers also stream large objects directly from the stream layer and cache frequently accessed data for efficiency.

3.4 Two Replication Engines

Before describing the stream and partition layers in detail, we first give a brief overview of the two replication engines in our system and their separate responsibilities.

Intra-Stamp Replication (stream layer) – This system provides *synchronous replication* and is focused on making sure all the data written into a stamp is kept durable within that stamp. It keeps enough replicas of the data across different nodes in different fault domains to keep data durable within the stamp in the face of disk, node, and rack failures. Intra-stamp replication is done completely by the stream layer and is on the critical path of the customer's write requests. Once a transaction has been replicated successfully with intra-stamp replication, success can be returned back to the customer.

Inter-Stamp Replication (partition layer) – This system provides *asynchronous replication* and is focused on replicating data across stamps. Inter-stamp replication is done in the background and is off the critical path of the customer's request. This replication is at the object level, where either the whole object is replicated or recent delta changes are replicated for a given account. Inter-stamp replication is used for (a) keeping a copy of an account's data in two locations for disaster recovery and (b) migrating an account's data between stamps. Inter-stamp replication is configured for an account by the location service and performed by the partition layer.

Inter-stamp replication is focused on replicating objects and the transactions applied to those objects, whereas intra-stamp replication is focused on replicating blocks of disk storage that are used to make up the objects.

We separated replication into intra-stamp and inter-stamp at these two different layers for the following reasons. Intra-stamp replication provides durability against hardware failures, which occur frequently in large scale systems, whereas inter-stamp replication provides geo-redundancy against geo-disasters, which

are rare. It is crucial to provide intra-stamp replication with low latency, since that is on the critical path of user requests; whereas the focus of inter-stamp replication is optimal use of network bandwidth between stamps while achieving an acceptable level of replication delay. They are different problems addressed by the two replication schemes.

Another reason for creating these two separate replication layers is the namespace each of these two layers has to maintain. Performing intra-stamp replication at the stream layer allows the amount of information that needs to be maintained to be scoped by the size of a single storage stamp. This focus allows all of the meta-state for intra-stamp replication to be cached in memory for performance (see Section 4), enabling WAS to provide fast replication with strong consistency by quickly committing transactions within a single stamp for customer requests. In contrast, the partition layer combined with the location service controls and understands the global object namespace across stamps, allowing it to efficiently replicate and maintain object state across data centers.

4. Stream Layer

The stream layer provides an internal interface used only by the partition layer. It provides a file system like namespace and API, except that all writes are append-only. It allows clients (the partition layer) to open, close, delete, rename, read, append to, and concatenate these large files, which are called streams. A stream is an ordered list of extent pointers, and an extent is a sequence of append blocks.

Figure 2 shows stream "//foo", which contains (pointers to) four extents (E1, E2, E3, and E4). Each extent contains a set of blocks that were appended to it. E1, E2 and E3 are sealed extents. It means that they can no longer be appended to; only the last extent in a stream (E4) can be appended to. If an application reads the data of the stream from beginning to end, it would get the block contents of the extents in the order of E1, E2, E3 and E4.

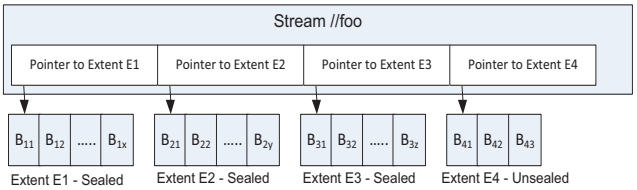

Figure 2: Example stream with four extents

In more detail these data concepts are:

Block – This is the minimum unit of data for writing and reading. A block can be up to *N* bytes (e.g. 4MB). Data is written (appended) as one or more concatenated blocks to an extent, where blocks do not have to be the same size. The client does an append in terms of blocks and controls the size of each block. A client read gives an offset to a stream or extent, and the stream layer reads as many blocks as needed at the offset to fulfill the length of the read. When performing a read, the entire contents of a block are read. This is because the stream layer stores its checksum validation at the block level, one checksum per block. The whole block is read to perform the checksum validation, and it is checked on every block read. In addition, all blocks in the system are validated against their checksums once every few days to check for data integrity issues.

Extent – Extents are the unit of replication in the stream layer, and the default replication policy is to keep three replicas within a storage stamp for an extent. Each extent is stored in an NTFS file and consists of a sequence of blocks. The target extent size used by the partition layer is 1GB. To store small objects, the partition layer appends many of them to the same extent and even in the same block; to store large TB-sized objects (Blobs), the object is broken up over many extents by the partition layer. The partition layer keeps track of what streams, extents, and byte offsets in the extents in which objects are stored as part of its index.

Streams – Every stream has a name in the hierarchical namespace maintained at the stream layer, and a stream looks like a big file to the partition layer. Streams are appended to and can be randomly read from. A stream is an ordered list of **pointers to extents** which is maintained by the Stream Manager. When the extents are concatenated together they represent the full contiguous address space in which the stream can be read in the order they were added to the stream. A new stream can be constructed by concatenating extents from existing streams, which is a fast operation since it just updates a list of pointers. Only the last extent in the stream can be appended to. All of the prior extents in the stream are immutable.

4.1 Stream Manager and Extent Nodes

The two main architecture components of the stream layer are the Stream Manager (SM) and Extent Node (EN) (shown in Figure 3).

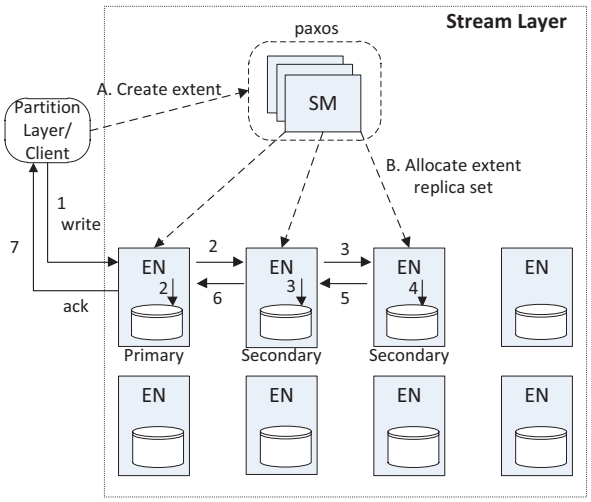

Figure 3: Stream Layer Architecture

Stream Manager (SM) – The SM keeps track of the stream namespace, what extents are in each stream, and the extent allocation across the Extent Nodes (EN). The SM is a standard Paxos cluster [13] as used in prior storage systems [3], and is off the critical path of client requests. The SM is responsible for (a) maintaining the stream namespace and state of all active streams and extents, (b) monitoring the health of the ENs, (c) creating and assigning extents to ENs, (d) performing the lazy re-replication of extent replicas that are lost due to hardware failures or unavailability, (e) garbage collecting extents that are no longer pointed to by any stream, and (f) scheduling the erasure coding of extent data according to stream policy (see Section 4.4).

The SM periodically polls (syncs) the state of the ENs and what extents they store. If the SM discovers that an extent is replicated on fewer than the expected number of ENs, a re-replication of the extent will lazily be created by the SM to regain the desired level of replication. For extent replica placement, the SM randomly chooses ENs across different fault domains, so that they are stored on nodes that will not have correlated failures due to power, network, or being on the same rack.

The SM does not know anything about blocks, just streams and extents. The SM is off the critical path of client requests and does not track each block append, since the total number of blocks can be huge and the SM cannot scale to track those. Since the stream and extent state is only tracked within a single stamp, the amount of state can be kept small enough to fit in the SM's memory. The only client of the stream layer is the partition layer, and the partition layer and stream layer are co-designed so that they will not use more than 50 million extents and no more than 100,000 streams for a single storage stamp given our current stamp sizes. This parameterization can comfortably fit into 32GB of memory for the SM.

Extent Nodes (EN) – Each extent node maintains the storage for a set of extent replicas assigned to it by the SM. An EN has *N* disks attached, which it completely controls for storing extent replicas and their blocks. An EN knows nothing about streams, and only deals with extents and blocks. Internally on an EN server, every extent on disk is a file, which holds data blocks and their checksums, and an index which maps extent offsets to blocks and their file location. Each extent node contains a view about the extents it owns and where the peer replicas are for a given extent. This view is a cache kept by the EN of the global state the SM keeps. ENs only talk to other ENs to replicate block writes (appends) sent by a client, or to create additional copies of an existing replica when told to by the SM. When an extent is no longer referenced by any stream, the SM garbage collects the extent and notifies the ENs to reclaim the space.

4.2 Append Operation and Sealed Extent

Streams can only be appended to; existing data cannot be modified. The append operations are atomic: either the entire data block is appended, or nothing is. Multiple blocks can be appended at once, as a single atomic "multi-block append" operation. The minimum read size from a stream is a single block. The "multi-block append" operation allows us to write a large amount of sequential data in a single append and to later perform small reads. The contract used between the client (partition layer) and the stream layer is that the multi-block append will occur atomically, and if the client never hears back for a request (due to failure) the client should retry the request (or seal the extent). This contract implies that the client needs to expect the same block to be appended more than once in face of timeouts and correctly deal with processing duplicate records. The partition layer deals with duplicate records in two ways (see Section 5 for details on the partition layer streams). For the metadata and commit log streams, all of the transactions written have a sequence number and duplicate records will have the same sequence number. For the row data and blob data streams, for duplicate writes, only the last write will be pointed to by the RangePartition data structures, so the prior duplicate writes will have no references and will be garbage collected later.

An extent has a target size, specified by the client (partition layer), and when it fills up to that size the extent is sealed at a block boundary, and then a new extent is added to the stream and appends continue into that new extent. Once an extent is **sealed** it can no longer be appended to. A sealed extent is immutable, and the stream layer performs certain optimizations on sealed extents like erasure coding cold extents. Extents in a stream do not have to be the same size, and they can be sealed anytime and can even grow arbitrarily large.

4.3 Stream Layer Intra-Stamp Replication

The stream layer and partition layer are co-designed to provide strong consistency at the object transaction level. The correctness of the partition layer providing strong consistency is built upon the following guarantees from the stream layer:

1. Once a record is appended and acknowledged back to the client, any later reads of that record from any replica will see the same data (the data is immutable).

2. Once an extent is sealed, any reads from any sealed replica will always see the same contents of the extent.

The data center, Fabric Controller, and WAS have security mechanisms in place to guard against malicious adversaries, so the stream replication does not deal with such threats. We consider faults ranging from disk and node errors to power failures, network issues, bit-flip and random hardware failures, as well as software bugs. These faults can cause data corruption; checksums are used to detect such corruption. The rest of the section discusses the intra-stamp replication scheme within this context.

4.3.1 Replication Flow

As shown in Figure 3, when a stream is first created (step A), the SM assigns three replicas for the first extent (one primary and two secondary) to three extent nodes (step B), which are chosen by the SM to randomly spread the replicas across different fault and upgrade domains while considering extent node usage (for load balancing). In addition, the SM decides which replica will be the primary for the extent. Writes to an extent are always performed from the client to the primary EN, and the primary EN is in charge of coordinating the write to two secondary ENs. The primary EN and the location of the three replicas never change for an extent while it is being appended to (while the extent is **unsealed**). Therefore, no leases are used to represent the primary EN for an extent, since the primary is always fixed while an extent is unsealed.

When the SM allocates the extent, the extent information is sent back to the client, which then knows which ENs hold the three replicas and which one is the primary. This state is now part of the stream's metadata information held in the SM and cached on the client. When the last extent in the stream that is being appended to becomes sealed, the same process repeats. The SM then allocates another extent, which now becomes the last extent in the stream, and all new appends now go to the new last extent for the stream.

For an extent, every append is replicated three times across the extent's replicas. A client sends all write requests to the primary EN, but it can read from any replica, even for unsealed extents. The append is sent to the primary EN for the extent by the client, and the primary is then in charge of (a) determining the offset of the append in the extent, (b) ordering (choosing the offset of) all of the appends if there are concurrent append requests to the same extent outstanding, (c) sending the append with its chosen offset to the two secondary extent nodes, and (d) only returning success for the append to the client after a successful append has occurred to disk for all three extent nodes. The sequence of steps during an append is shown in Figure 3 (labeled with numbers). Only when all of the writes have succeeded for all three replicas will the primary EN then respond to the client that the append was a success. If there are multiple outstanding appends to the same extent, the primary EN will respond success in the order of their offset (commit them in order) to the clients. As appends commit in order for a replica, the last append position is considered to be the current **commit length** of the replica. We ensure that the bits are the same between all replicas by the fact that the primary EN for an extent never changes, it always picks the offset for appends,

appends for an extent are committed in order, and how extents are sealed upon failures (discussed in Section 4.3.2).

When a stream is opened, the metadata for its extents is cached at the client, so the client can go directly to the ENs for reading and writing without talking to the SM until the next extent needs to be allocated for the stream. If during writing, one of the replica's ENs is not reachable or there is a disk failure for one of the replicas, a write failure is returned to the client. The client then contacts the SM, and the extent that was being appended to is sealed by the SM at its current commit length (see Section 4.3.2). At this point the sealed extent can no longer be appended to. The SM will then allocate a new extent with replicas on different (available) ENs, which makes it now the last extent of the stream. The information for this new extent is returned to the client. The client then continues appending to the stream with its new extent. This process of sealing by the SM and allocating the new extent is done on average within 20ms. A key point here is that the client can continue appending to a stream as soon as the new extent has been allocated, and it does not rely on a specific node to become available again.

For the newly sealed extent, the SM will create new replicas to bring it back to the expected level of redundancy in the background if needed.

4.3.2 Sealing

From a high level, the SM coordinates the sealing operation among the ENs; it determines the commit length of the extent used for sealing based on the commit length of the extent replicas. Once the sealing is done, the commit length will never change again.

To seal an extent, the SM asks all three ENs their current length. During sealing, either all replicas have the same length, which is the simple case, or a given replica is longer or shorter than another replica for the extent. This latter case can only occur during an append failure where some but not all of the ENs for the replica are available (i.e., some of the replicas get the append block, but not all of them). We guarantee that the SM will seal the extent even if the SM may not be able to reach all the ENs involved. When sealing the extent, the SM will choose the smallest commit length based on the available ENs it can talk to. This will not cause data loss since the primary EN will not return success unless all replicas have been written to disk for all three ENs. This means the smallest commit length is sure to contain all the writes that have been acknowledged to the client. In addition, it is also fine if the final length contains blocks that were never acknowledged back to the client, since the client (partition layer) correctly deals with these as described in Section 4.2. During the sealing, all of the extent replicas that were reachable by the SM are sealed to the commit length chosen by the SM.

Once the sealing is done, the commit length of the extent will never be changed. If an EN was not reachable by the SM during the sealing process but later becomes reachable, the SM will force the EN to synchronize the given extent to the chosen commit length. This ensures that once an extent is sealed, all its available replicas (the ones the SM can eventually reach) are bitwise identical.

4.3.3 Interaction with Partition Layer

An interesting case is when, due to network partitioning, a client (partition server) is still able to talk to an EN that the SM could not talk to during the sealing process. This section explains how the partition layer handles this case.

The partition layer has two different read patterns:

1. **Read records at known locations.** The partition layer uses two types of data streams (row and blob). For these streams, it always reads at specific locations (extent+offset, length). More importantly, the partition layer will only read these two streams using the location information returned from a previous successful append at the stream layer. That will only occur if the append was successfully committed to all three replicas. The replication scheme guarantees such reads always see the same data.

2. **Iterate all records sequentially in a stream on partition load.** Each partition has two additional streams (metadata and commit log). These are the only streams that the partition layer will read sequentially from a starting point to the very last record of a stream. This operation only occurs when the partition is loaded (explained in Section 5). The partition layer ensures that no useful appends from the partition layer will happen to these two streams during partition load. Then the partition and stream layer together ensure that the same sequence of records is returned on partition load.

At the start of a partition load, the partition server sends a "check for commit length" to the primary EN of the last extent of these two streams. This checks whether all the replicas are available and that they all have the same length. If not, the extent is sealed and reads are only performed, during partition load, against a replica sealed by the SM. This ensures that the partition load will see all of its data and the exact same view, even if we were to repeatedly load the same partition reading from different sealed replicas for the last extent of the stream.

4.4 Erasure Coding Sealed Extents

To reduce the cost of storage, WAS erasure codes sealed extents for Blob storage. WAS breaks an extent into N roughly equal sized fragments at block boundaries. Then, it adds M error correcting code fragments using Reed-Solomon for the erasure coding algorithm [19]. As long as it does not lose more than M fragments (across the data fragments + code fragments), WAS can recreate the full extent.

Erasure coding sealed extents is an important optimization, given the amount of data we are storing. It reduces the cost of storing data from three full replicas within a stamp, which is three times the original data, to only 1.3x – 1.5x the original data, depending on the number of fragments used. In addition, erasure coding actually increases the durability of the data when compared to keeping three replicas within a stamp.

4.5 Read Load-Balancing

When reads are issued for an extent that has three replicas, they are submitted with a "deadline" value which specifies that the read should not be attempted if it cannot be fulfilled within the deadline. If the EN determines the read cannot be fulfilled within the time constraint, it will immediately reply to the client that the deadline cannot be met. This mechanism allows the client to select a different EN to read that data from, likely allowing the read to complete faster.

This method is also used with erasure coded data. When reads cannot be serviced in a timely manner due to a heavily loaded spindle to the data fragment, the read may be serviced faster by doing a reconstruction rather than reading that data fragment. In this case, reads (for the range of the fragment needed to satisfy the client request) are issued to all fragments of an erasure coded extent, and the first N responses are used to reconstruct the desired fragment.

4.6 Spindle Anti-Starvation

Many hard disk drives are optimized to achieve the highest possible throughput, and sacrifice fairness to achieve that goal. They tend to prefer reads or writes that are sequential. Since our system contains many streams that can be very large, we observed in developing our service that some disks would lock into servicing large pipelined reads or writes while starving other operations. On some disks we observed this could lock out non-sequential IO for as long as 2300 milliseconds. To avoid this problem we avoid scheduling new IO to a spindle when there is over 100ms of expected pending IO already scheduled or when there is any pending IO request that has been scheduled but not serviced for over 200ms. Using our own custom IO scheduling allows us to achieve fairness across reads/writes at the cost of slightly increasing overall latency on some sequential requests.

4.7 Durability and Journaling

The durability contract for the stream layer is that when data is acknowledged as written by the stream layer, there must be at least three durable copies of the data stored in the system. This contract allows the system to maintain data durability even in the face of a cluster-wide power failure. We operate our storage system in such a way that all writes are made durable to power safe storage before they are acknowledged back to the client.

As part of maintaining the durability contract while still achieving good performance, an important optimization for the stream layer is that on each extent node we reserve a whole disk drive or SSD as a **journal drive** for all writes into the extent node. The journal drive [11] is dedicated solely for writing a single sequential journal of data, which allows us to reach the full write throughput potential of the device. When the partition layer does a stream append, the data is written by the primary EN while in parallel sent to the two secondaries to be written. When each EN performs its append, it (a) writes all of the data for the append to the journal drive and (b) queues up the append to go to the data disk where the extent file lives on that EN. Once either succeeds, success can be returned. If the journal succeeds first, the data is also buffered in memory while it goes to the data disk, and any reads for that data are served from memory until the data is on the data disk. From that point on, the data is served from the data disk. This also enables the combining of contiguous writes into larger writes to the data disk, and better scheduling of concurrent writes and reads to get the best throughput. It is a tradeoff for good latency at the cost of an extra write off the critical path.

Even though the stream layer is an append-only system, we found that adding a journal drive provided important benefits, since the appends do not have to contend with reads going to the data disk in order to commit the result back to the client. The journal allows the append times from the partition layer to have more consistent and lower latencies. Take for example the partition layer's commit log stream, where an append is only as fast as the slowest EN for the replicas being appended to. For small appends to the commit log stream without journaling we saw an average end-to-end stream append latency of 30ms. With journaling we see an average append latency of 6ms. In addition, the variance of latencies decreased significantly.

5. Partition Layer

The partition layer stores the different types of objects and understands what a transaction means for a given object type (Blob, Table, or Queue). The partition layer provides the (a) data model for the different types of objects stored, (b) logic and semantics to process the different types of objects, (c) massively scalable namespace for the objects, (d) load balancing to access objects across the available partition servers, and (e) transaction ordering and strong consistency for access to objects.

5.1 Partition Layer Data Model

The partition layer provides an important internal data structure called an Object Table (OT). An OT is a massive table which can grow to several petabytes. Object Tables are dynamically broken up into RangePartitions (based on traffic load to the table) and spread across Partition Servers (Section 5.2) in a stamp. A **RangePartition** is a contiguous range of rows in an OT from a given low-key to a high-key. All RangePartitions for a given OT are non-overlapping, and every row is represented in some RangePartition.

The following are the Object Tables used by the partition layer. The Account Table stores metadata and configuration for each storage account assigned to the stamp. The Blob Table stores all blob objects for all accounts in the stamp. The Entity Table stores all entity rows for all accounts in the stamp; it is used for the public Windows Azure Table data abstraction. The Message Table stores all messages for all accounts' queues in the stamp. The Schema Table keeps track of the schema for all OTs. The Partition Map Table keeps track of the current RangePartitions for all Object Tables and what partition server is serving each RangePartition. This table is used by the Front-End servers to route requests to the corresponding partition servers.

Each of the above OTs has a fixed schema stored in the Schema Table. The primary key for the Blob Table, Entity Table, and Message Table consists of three properties: AccountName, PartitionName, and ObjectName. These properties provide the indexing and sort order for those Object Tables.

5.1.1 Supported Data Types and Operations

The property types supported for an OT's schema are the standard simple types (bool, binary, string, DateTime, double, GUID, int32, int64). In addition, the system supports two special types – DictionaryType and BlobType. The DictionaryType allows for flexible properties (i.e., without a fixed schema) to be added to a row at any time. These flexible properties are stored inside of the dictionary type as (name, type, value) tuples. From a data access standpoint, these flexible properties behave like first-order properties of the row and are queryable just like any other property in the row. The BlobType is a special property used to store large amounts of data and is currently used only by the Blob Table. BlobType avoids storing the blob data bits with the row properties in the "row data stream". Instead, the blob data bits are stored in a separate "blob data stream" and a pointer to the blob's data bits (list of "extent + offset, length" pointers) is stored in the BlobType's property in the row. This keeps the large data bits separated from the OT's queryable row property values stored in the row data stream.

OTs support standard operations including insert, update, and delete operations on rows as well as query/get operations. In addition, OTs allows batch transactions across rows with the same PartitionName value. The operations in a single batch are committed as a single transaction. Finally, OTs provide snapshot isolation to allow read operations to happen concurrently with writes.

5.2 Partition Layer Architecture

The partition layer has three main architectural components as shown in Figure 4: a Partition Manager (PM), Partition Servers (PS), and a Lock Service.

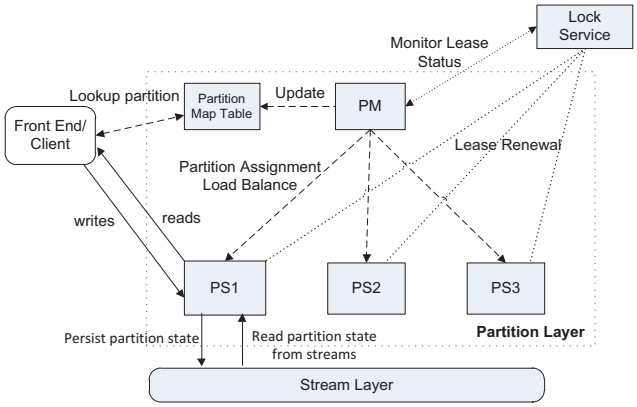

Figure 4: Partition Layer Architecture

Partition Manager (PM) – Responsible for keeping track of and splitting the massive Object Tables into RangePartitions and assigning each RangePartition to a Partition Server to serve access to the objects. The PM splits the Object Tables into *N* RangePartitions in each stamp, keeping track of the current RangePartition breakdown for each OT and to which partition servers they are assigned. The PM stores this assignment in the Partition Map Table. The PM ensures that each RangePartition is assigned to exactly one active partition server at any time, and that two RangePartitions do not overlap. It is also responsible for load balancing RangePartitions among partition servers. Each stamp has multiple instances of the PM running, and they all contend for a leader lock that is stored in the Lock Service (see below). The PM with the lease is the active PM controlling the partition layer.

Partition Server (PS) – A partition server is responsible for serving requests to a set of RangePartitions assigned to it by the PM. The PS stores all the persistent state of the partitions into streams and maintains a memory cache of the partition state for efficiency. The system guarantees that no two partition servers can serve the same RangePartition at the same time by using leases with the Lock Service. This allows the PS to provide strong consistency and ordering of concurrent transactions to objects for a RangePartition it is serving. A PS can concurrently serve multiple RangePartitions from different OTs. In our deployments, a PS serves on average ten RangePartitions at any time.

Lock Service – A Paxos Lock Service [3,13] is used for leader election for the PM. In addition, each PS also maintains a lease with the lock service in order to serve partitions. We do not go into the details of the PM leader election, or the PS lease management, since the concepts used are similar to those described in the Chubby Lock [3] paper.

On partition server failure, all *N* RangePartitions served by the failed PS are assigned to available PSs by the PM. The PM will choose *N* (or fewer) partition servers, based on the load on those servers. The PM will assign a RangePartition to a PS, and then update the Partition Map Table specifying what partition server is serving each RangePartition. This allows the Front-End layer to find the location of RangePartitions by looking in the Partition Map Table (see Figure 4). When the PS gets a new assignment it will start serving the new RangePartitions for as long as the PS holds its partition server lease.

5.3 RangePartition Data Structures

A PS serves a RangePartition by maintaining a set of in-memory data structures and a set of persistent data structures in streams.

5.3.1 Persistent Data Structure

A RangePartition uses a Log-Structured Merge-Tree [17,4] to maintain its persistent data. Each Object Table's RangePartition consists of its own set of streams in the stream layer, and the streams belong solely to a given RangePartition, though the underlying extents can be pointed to by multiple streams in different RangePartitions due to RangePartition splitting. The following are the set of streams that comprise each RangePartition (shown in Figure 5):

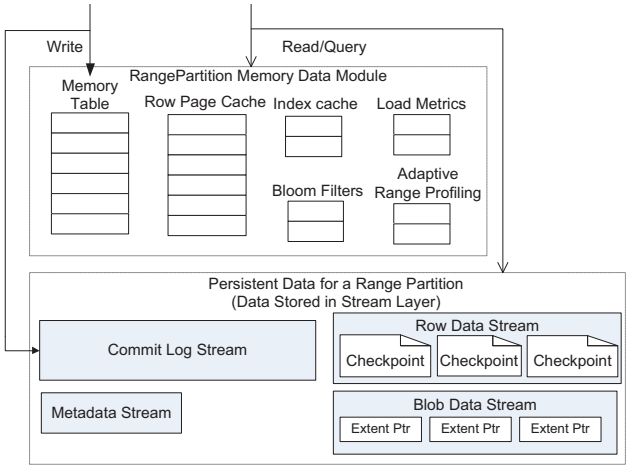

Figure 5: RangePartition Data Structures

Metadata Stream – The metadata stream is the root stream for a RangePartition. The PM assigns a partition to a PS by providing the name of the RangePartition's metadata stream. The metadata stream contains enough information for the PS to load a RangePartition, including the name of the commit log stream and data streams for that RangePartition, as well as pointers (extent+offset) into those streams for where to start operating in those streams (e.g., where to start processing in the commit log stream and the root of the index for the row data stream). The PS serving the RangePartition also writes in the metadata stream the status of outstanding split and merge operations that the RangePartition may be involved in.

Commit Log Stream – Is a commit log used to store the recent insert, update, and delete operations applied to the RangePartition since the last checkpoint was generated for the RangePartition.

Row Data Stream – Stores the checkpoint row data and index for the RangePartition.

Blob Data Stream – Is only used by the Blob Table to store the blob data bits.

Each of the above is a separate stream in the stream layer owned by an Object Table's RangePartition.

Each RangePartition in an Object Table has only one data stream, except the Blob Table. A RangePartition in the Blob Table has a "row data stream" for storing its row checkpoint data (the blob index), and a separate "blob data stream" for storing the blob data bits for the special BlobType described earlier.

5.3.2 In-Memory Data Structures

A partition server maintains the following in-memory components as shown in Figure 5:

Memory Table – This is the in-memory version of the commit log for a RangePartition, containing all of the recent updates that have not yet been checkpointed to the row data stream. When a

lookup occurs the memory table is checked to find recent updates to the RangePartition.

Index Cache – This cache stores the checkpoint indexes of the row data stream. We separate this cache out from the row data cache to make sure we keep as much of the main index cached in memory as possible for a given RangePartition.

Row Data Cache – This is a memory cache of the checkpoint row data pages. The row data cache is read-only. When a lookup occurs, both the row data cache and the memory table are checked, giving preference to the memory table.

Bloom Filters – If the data is not found in the memory table or the row data cache, then the index/checkpoints in the data stream need to be searched. It can be expensive to blindly examine them all. Therefore a bloom filter is kept for each checkpoint, which indicates if the row being accessed *may* be in the checkpoint.

We do not go into further details about these components, since these are similar to those in [17,4].

5.4 Data Flow

When the PS receives a write request to the RangePartition (e.g., insert, update, delete), it appends the operation into the commit log, and then puts the newly changed row into the memory table. Therefore, all the modifications to the partition are recorded persistently in the commit log, and also reflected in the memory table. At this point success can be returned back to the client (the FE servers) for the transaction. When the size of the memory table reaches its threshold size or the size of the commit log stream reaches its threshold, the partition server will write the contents of the memory table into a checkpoint stored persistently in the row data stream for the RangePartition. The corresponding portion of the commit log can then be removed. To control the total number of checkpoints for a RangePartition, the partition server will periodically combine the checkpoints into larger checkpoints, and then remove the old checkpoints via garbage collection.

For the Blob Table's RangePartitions, we also store the Blob data bits directly into the commit log stream (to minimize the number of stream writes for Blob operations), but those data bits are not part of the row data so they are not put into the memory table. Instead, the BlobType property for the row tracks the location of the Blob data bits (extent+offset, length). During checkpoint, the extents that would be removed from the commit log are instead concatenated to the RangePartition's Blob data stream. Extent concatenation is a fast operation provided by the stream layer since it consists of just adding pointers to extents at the end of the Blob data stream without copying any data.

A PS can start serving a RangePartition by "loading" the partition. Loading a partition involves reading the metadata stream of the RangePartition to locate the active set of checkpoints and replaying the transactions in the commit log to rebuild the in-memory state. Once these are done, the PS has the up-to-date view of the RangePartition and can start serving requests.

5.5 RangePartition Load Balancing

A critical part of the partition layer is breaking these massive Object Tables into RangePartitions and automatically load balancing them across the partition servers to meet their varying traffic demands.

The PM performs three operations to spread load across partition servers and control the total number of partitions in a stamp:

Load Balance – This operation identifies when a given PS has too much traffic and reassigns one or more RangePartitions to less loaded partition servers.

Split – This operation identifies when a single RangePartition has too much load and splits the RangePartition into two or more smaller and disjoint RangePartitions, then load balances (reassigns) them across two or more partition servers.

Merge – This operation merges together cold or lightly loaded RangePartitions that together form a contiguous key range within their OT. Merge is used to keep the number of RangePartitions within a bound proportional to the number of partition servers in a stamp.

WAS keeps the total number of partitions between a low watermark and a high watermark (typically around ten times the partition server count within a stamp). At equilibrium, the partition count will stay around the low watermark. If there are unanticipated traffic bursts that concentrate on a single RangePartition, it will be split to spread the load. When the total RangePartition count is approaching the high watermark, the system will increase the merge rate to eventually bring the RangePartition count down towards the low watermark. Therefore, the number of RangePartitions for each OT changes dynamically based upon the load on the objects in those tables.

Having a high watermark of RangePartitions ten times the number of partition servers (a storage stamp has a few hundred partition servers) was chosen based on how big we can allow the stream and extent metadata to grow for the SM, and still completely fit the metadata in memory for the SM. Keeping many more RangePartitions than partition servers enables us to quickly distribute a failed PS or rack's load across many other PSs. A given partition server can end up serving a single extremely hot RangePartition, tens of lightly loaded RangePartitions, or a mixture in-between, depending upon the current load to the RangePartitions in the stamp. The number of RangePartitions for the Blob Table vs. Entity Table vs. Message Table depends upon the load on the objects in those tables and is continuously changing within a storage stamp based upon traffic.

For each stamp, we typically see 75 splits and merges and 200 RangePartition load balances per day.

5.5.1 Load Balance Operation Details

We track the load for each RangePartition as well as the overall load for each PS. For both of these we track (a) transactions/second, (b) average pending transaction count, (c) throttling rate, (d) CPU usage, (e) network usage, (f) request latency, and (g) data size of the RangePartition. The PM maintains heartbeats with each PS. This information is passed back to the PM in responses to the heartbeats. If the PM sees a RangePartition that has too much load based upon the metrics, then it will decide to split the partition and send a command to the PS to perform the split. If instead a PS has too much load, but no individual RangePartition seems to be too highly loaded, the PM will take one or more RangePartitions from the PS and reassign them to a more lightly loaded PS.

To load balance a RangePartition, the PM sends an offload command to the PS, which will have the RangePartition write a current checkpoint before offloading it. Once complete, the PS acks back to the PM that the offload is done. The PM then assigns the RangePartition to its new PS and updates the Partition Map Table to point to the new PS. The new PS loads and starts serving traffic for the RangePartition. The loading of the RangePartition on the new PS is very quick since the commit log is small due to the checkpoint prior to the offload.

5.5.2 Split Operation

WAS splits a RangePartition due to too much load as well as the size of its row or blob data streams. If the PM identifies either situation, it tells the PS serving the RangePartition to split based upon load or size. The PM makes the decision to split, but the PS chooses the key (AccountName, PartitionName) where the partition will be split. To split based upon size, the RangePartition maintains the total size of the objects in the partition and the split key values where the partition can be approximately halved in size, and the PS uses that to pick the key for where to split. If the split is based on load, the PS chooses the key based upon Adaptive Range Profiling [16]. The PS adaptively tracks which key ranges in a RangePartition have the most load and uses this to determine on what key to split the RangePartition.

To split a RangePartition (B) into two new RangePartitions (C,D), the following steps are taken.

1. The PM instructs the PS to split B into C and D.

2. The PS in charge of B checkpoints B, then stops serving traffic briefly during step 3 below.

3. The PS uses a special stream operation "MultiModify" to take each of B's streams (metadata, commit log and data) and creates new sets of streams for C and D respectively with the same extents in the same order as in B. This step is very fast, since a stream is just a list of pointers to extents. The PS then appends the new partition key ranges for C and D to their metadata streams.

4. The PS starts serving requests to the two new partitions C and D for their respective disjoint PartitionName ranges.

5. The PS notifies the PM of the split completion, and the PM updates the Partition Map Table and its metadata information accordingly. The PM then moves one of the split partitions to a different PS.

5.5.3 Merge Operation

To merge two RangePartitions, the PM will choose two RangePartitions C and D with adjacent PartitionName ranges that have low traffic. The following steps are taken to merge C and D into a new RangePartition E.

1. The PM moves C and D so that they are served by the same PS. The PM then tells the PS to merge (C,D) into E.

2. The PS performs a checkpoint for both C and D, and then briefly pauses traffic to C and D during step 3.

3. The PS uses the MultiModify stream command to create a new commit log and data streams for E. Each of these streams is the concatenation of all of the extents from their respective streams in C and D. This merge means that the extents in the new commit log stream for E will be all of C's extents in the order they were in C's commit log stream followed by all of D's extents in their original order. This layout is the same for the new row and Blob data stream(s) for E.

4. The PS constructs the metadata stream for E, which contains the names of the new commit log and data stream, the combined key range for E, and pointers (extent+offset) for the start and end of the commit log regions in E's commit log derived from C and D, as well as the root of the data index in E's data streams.

5. At this point, the new metadata stream for E can be correctly loaded, and the PS starts serving the newly merged RangePartition E.

6. The PM then updates the Partition Map Table and its metadata information to reflect the merge.

5.6 Partition Layer Inter-Stamp Replication

Thus far we have talked about an AccountName being associated (via DNS) to a single location and storage stamp, where all data access goes to that stamp. We call this the **primary stamp** for an account. An account actually has one or more **secondary stamps** assigned to it by the Location Service, and this primary/secondary stamp information tells WAS to perform inter-stamp replication for this account from the primary stamp to the secondary stamp(s).

One of the main scenarios for inter-stamp replication is to geo-replicate an account's data between two data centers for disaster recovery. In this scenario, a primary and secondary location is chosen for the account. Take, for example, an account, for which we want the primary stamp (P) to be located in US South and the secondary stamp (S) to be located in US North. When provisioning the account, the LS will choose a stamp in each location and register the AccountName with both stamps such that the US South stamp (P) takes live traffic and the US North stamp (S) will take only inter-stamp replication (also called geo-replication) traffic from stamp P for the account. The LS updates DNS to have hostname **AccountName**.service.core.windows.net point to the storage stamp P's VIP in US South. When a write comes into stamp P for the account, the change is fully replicated within that stamp using intra-stamp replication at the stream layer then success is returned to the client. After the update has been committed in stamp P, the partition layer in stamp P will asynchronously geo-replicate the change to the secondary stamp S using inter-stamp replication. When the change arrives at stamp S, the transaction is applied in the partition layer and this update fully replicates using intra-stamp replication within stamp S.

Since the inter-stamp replication is done asynchronously, recent updates that have not been inter-stamp replicated can be lost in the event of disaster. In production, changes are geo-replicated and committed on the secondary stamp within 30 seconds on average after the update was committed on the primary stamp.

Inter-stamp replication is used for both account geo-replication and migration across stamps. For disaster recovery, we may need to perform an abrupt failover where recent changes may be lost, but for migration we perform a clean failover so there is no data loss. In both failover scenarios, the Location Service makes an active secondary stamp for the account the new primary and switches DNS to point to the secondary stamp's VIP. Note that the URI used to access the object does not change after failover. This allows the existing URIs used to access Blobs, Tables and Queues to continue to work after failover.

6. Application Throughput

For our cloud offering, customers run their applications as a tenant (service) on VMs. For our platform, we separate computation and storage into their own stamps (clusters) within a data center since this separation allows each to scale independently and control their own load balancing. Here we examine the performance of a customer application running from their hosted service on VMs in the same data center as where their account data is stored. Each VM used is an extra-large VM with full control of the entire compute node and a 1Gbps NIC. The results were gathered on live shared production stamps with internal and external customers.

Figure 6 shows the WAS Table operation throughput in terms of the entities per second (*y*-axis) for 1-16 VMs (*x*-axis) performing random 1KB single entity get and put requests against a single 100GB Table. It also shows batch inserts of 100 entities at a time – a common way applications insert groups of entities into a WAS Table. Figure 7 shows the throughput in megabytes per second (*y*-axis) for randomly getting and putting 4MB blobs vs. the number of VMs used (*x*-axis). All of the results are for a single storage account.

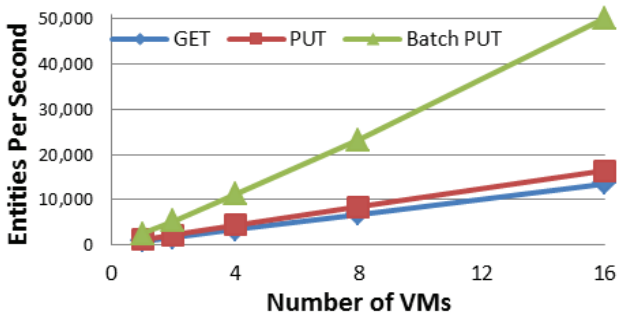

Figure 6 Table Entity Throughput for 1-16 VMs

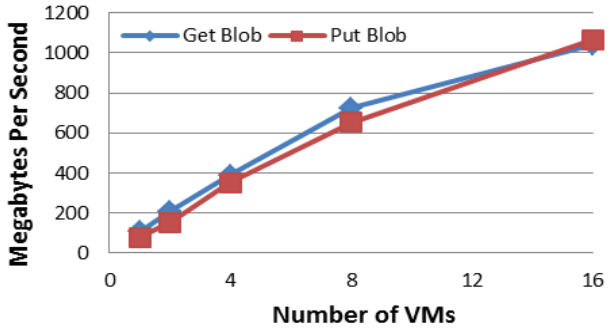

Figure 7: Blob Throughput for 1-16 VMs

These results show a linear increase in scale is achieved for entities/second as the application scales out the amount of computing resources it uses for accessing WAS Tables. For Blobs, the throughput scales linearly up to eight VMs, but tapers off as the aggregate throughput reaches the network capacity on the client side where the test traffic was generated. The results show that, for Table operations, batch puts offer about three times more throughput compared to single entity puts. That is because the batch operation significantly reduces the number of network roundtrips and requires fewer stream writes. In addition, the Table read operations have slightly lower throughput than write operations. This difference is due to the particular access pattern of our experiment, which randomly accesses a large key space on a large data set, minimizing the effect of caching. Writes on the other hand always result in sequential writes to the journal.

7. Workload Profiles

Usage patterns for cloud-based applications can vary significantly. Section 1 already described a near-real time ingestion engine to provide Facebook and Twitter search for Bing. In this section we describe a few additional internal services using WAS, and give some high-level metrics of their usage.

The XBox GameSaves service was announced at E3 this year and will provide a new feature in Fall 2011 for providing saved game data into the cloud for millions of XBox users. This feature will enable subscribed users to upload their game progress into the WAS cloud storage service, which they can then access from any XBox console they sign into. The backing storage for this feature leverages Blob and Table storage.

The XBox Telemetry service stores console-generated diagnostics and telemetry information for later secure retrieval and offline processing. For example, various Kinect related features running on Xbox 360 generate detailed usage files which are uploaded to the cloud to analyze and improve the Kinect experience based on customer opt-in. The data is stored directly into Blobs, and Tables are used to maintain metadata information about the files. Queues are used to coordinate the processing and the cleaning up of the Blobs.

Microsoft's Zune backend uses Windows Azure for media file storage and delivery, where files are stored as Blobs.

Table 1 shows the relative breakdown among Blob, Table, and Queue usage across all (All) services (internal and external) using WAS as well as for the services described above. The table shows the breakdown of requests, capacity usage, and ingress and egress traffic for Blobs, Tables and Queues.

Notice that, the percentage of requests for all services shows that about 17.9% of all requests are Blob requests, 46.88% of the requests are Table operations and 35.22% are Queue requests for all services using WAS. But in terms of capacity, 70.31% of capacity is in Blobs, 29.68% of capacity is used by Tables, and 0.01% used by Queues. "%Ingress" is the percentage breakdown of incoming traffic (bytes) among Blob, Table, and Queue; "%Egress" is the same for outbound traffic (bytes). The results show that different customers have very different usage patterns. In term of capacity usage, some customers (e.g., Zune and Xbox GameSaves) have mostly unstructured data (such as media files) and put those into Blobs, whereas other customers like Bing and XBox Telemetry that have to index a lot of data have a significant amount of structured data in Tables. Queues use very little space compared to Blobs and Tables, since they are primarily used as a communication mechanism instead of storing data over a long period of time.

Table 1: Usage Comparison for (Blob/Table/Queue)

		%Requests	%Capacity	%Ingress	%Egress
All	Blob	17.9	70.31	48.28	66.17
	Table	46.88	29.68	49.61	33.07
	Queue	35.22	0.01	2.11	0.76
Bing	Blob	0.46	60.45	16.73	29.11
	Table	98.48	39.55	83.14	70.79
	Queue	1.06	0	0.13	0.1
XBox GameSaves	Blob	99.68	99.99	99.84	99.88
	Table	0.32	0.01	0.16	0.12
	Queue	0	0	0	0
XBox Telemetry	Blob	26.78	19.57	50.25	11.26
	Table	44.98	80.43	49.25	88.29
	Queue	28.24	0	0.5	0.45
Zune	Blob	94.64	99.9	98.22	96.21
	Table	5.36	0.1	1.78	3.79
	Queue	0	0	0	0

8. Design Choices and Lessons Learned

Here, we discuss a few of our WAS design choices and relate some of the lessons we have learned thus far.

Scaling Computation Separate from Storage – Early on we decided to separate customer VM-based computation from storage for Windows Azure. Therefore, nodes running a customer's

service code are separate from nodes providing their storage. As a result, we can scale our supply of computation cores and storage independently to meet customer demand in a given data center. This separation also provides a layer of isolation between compute and storage given its multi-tenancy usage, and allows both of the systems to load balance independently.

Given this decision, our goal from the start has been to allow computation to efficiently access storage with high bandwidth without the data being on the same node or even in the same rack. To achieve this goal we are in the process of moving towards our next generation data center networking architecture [10], which flattens the data center networking topology and provides full bisection bandwidth between compute and storage.

Range Partitions vs. Hashing – We decided to use range-based partitioning/indexing instead of hash-based indexing (where the objects are assigned to a server based on the hash values of their keys) for the partition layer's Object Tables. One reason for this decision is that range-based partitioning makes performance isolation easier since a given account's objects are stored together within a set of RangePartitions (which also provides efficient object enumeration). Hash-based schemes have the simplicity of distributing the load across servers, but lose the locality of objects for isolation and efficient enumeration. The range partitioning allows WAS to keep a customer's objects together in their own set of partitions to throttle and isolate potentially abusive accounts.

For these reasons, we took the range-based approach and built an automatic load balancing system (Section 5.5) to spread the load dynamically according to user traffic by splitting and moving partitions among servers.

A downside of range partitioning is scaling out access to sequential access patterns. For example, if a customer is writing all of their data to the very end of a table's key range (e.g., insert key 2011-06-30:12:00:00, then key 2011-06-30:12:00:02, then key 2011-06:30-12:00:10), all of the writes go to the very last RangePartition in the customer's table. This pattern does not take advantage of the partitioning and load balancing our system provides. In contrast, if the customer distributes their writes across a large number of PartitionNames, the system can quickly split the table into multiple RangePartitions and spread them across different servers to allow performance to scale linearly with load (as shown in Figure 6). To address this sequential access pattern for RangePartitions, a customer can always use hashing or bucketing for the PartitionName, which avoids the above sequential access pattern issue.

Throttling/Isolation – At times, servers become overloaded by customer requests. A difficult problem was identifying which storage accounts should be throttled when this happens and making sure well-behaving accounts are not affected.

Each partition server keeps track of the request rate for AccountNames and PartitionNames. Because there are a large number of AccountNames and PartitionNames it may not be practical to keep track of them all. The system uses a Sample-Hold algorithm [7] to track the request rate history of the top *N* busiest AccountNames and PartitionNames. This information is used to determine whether an account is well-behaving or not (e.g., whether the traffic backs off when it is throttled). If a server is getting overloaded, it uses this information to selectively throttle the incoming traffic, targeting accounts that are causing the issue. For example, a PS computes a throttling probability of the incoming requests for each account based on the request rate history for the account (those with high request rates will have a larger probability being throttled, whereas accounts with little traffic will not). In addition, based on the request history at the AccountName and PartitionName levels, the system determines whether the account has been well-behaving. Load balancing will try to keep the servers within an acceptable load, but when access patterns cannot be load balanced (e.g., high traffic to a single PartitionName, high sequential access traffic, repetitive sequential scanning, etc.), the system throttles requests of such traffic patterns when they are too high.

Automatic Load Balancing – We found it crucial to have efficient automatic load balancing of partitions that can quickly adapt to various traffic conditions. This enables WAS to maintain high availability in this multi-tenancy environment as well as deal with traffic spikes to a single user's storage account. Gathering the adaptive profile information, discovering what metrics are most useful under various traffic conditions, and tuning the algorithm to be smart enough to effectively deal with different traffic patterns we see in production were some of the areas we spent a lot of time working on before achieving a system that works well for our multi-tenancy environment.

We started with a system that used a single number to quantify "load" on each RangePartition and each server. We first tried the product of request latency and request rate to represent the load on a PS and each RangePartition. This product is easy to compute and reflects the load incurred by the requests on the server and partitions. This design worked well for the majority of the load balancing needs (moving partitions around), but it did not correctly capture high CPU utilization that can occur during scans or high network utilization. Therefore, we now take into consideration request, CPU, and network loads to guide load balancing. However, these metrics are not sufficient to correctly guide splitting decisions.

For splitting, we introduced separate mechanisms to trigger splits of partitions, where we collect hints to find out whether some partitions are reaching their capacity across several metrics. For example, we can trigger partition splits based on request throttling, request timeouts, the size of a partition, etc. Combining split triggers and the load balancing allows the system to quickly split and load balance hot partitions across different servers.

From a high level, the algorithm works as follows. Every *N* seconds (currently 15 seconds) the PM sorts all RangePartitions based on each of the split triggers. The PM then goes through each partition, looking at the detailed statistics to figure out if it needs to be split using the metrics described above (load, throttling, timeouts, CPU usage, size, etc.). During this process, the PM picks a small number to split for this quantum, and performs the split action on those.

After doing the split pass, the PM sorts all of the PSs based on each of the load balancing metrics - request load, CPU load and network load. It then uses this to identify which PSs are overloaded versus lightly loaded. The PM then chooses the PSs that are heavily loaded and, if there was a recent split from the prior split pass, the PM will offload one of those RangePartitions to a lightly loaded server. If there are still highly loaded PSs (without a recent split to offload), the PM offloads RangePartitions from them to the lightly loaded PSs.

The core load balancing algorithm can be dynamically "swapped out" via configuration updates. WAS includes scripting language support that enables customizing the load balancing logic, such as defining how a partition split can be triggered based on different system metrics. This support gives us flexibility to fine-tune the load balancing algorithm at runtime as well as try new algorithms according to various traffic patterns observed.

Separate Log Files per RangePartition – Performance isolation for storage accounts is critical in a multi-tenancy environment. This requirement is one of the reasons we used separate log streams for each RangePartition, whereas BigTable [4] uses a single log file across all partitions on the same server. Having separate log files enables us to isolate the load time of a RangePartition to just the recent object updates in that RangePartition.

Journaling – When we originally released WAS, it did not have journaling. As a result, we experienced many hiccups with read/writes contending with each other on the same drive, noticeably affecting performance. We did not want to write to two log files (six replicas) like BigTable [4] due to the increased network traffic. We also wanted a way to optimize small writes, especially since we wanted separate log files per RangePartition. These requirements led us to the journal approach with a single log file per RangePartition. We found this optimization quite effective in reducing the latency and providing consistent performance.

Append-only System – Having an append-only system and sealing an extent upon failure have greatly simplified the replication protocol and handling of failure scenarios. In this model, the data is never overwritten once committed to a replica, and, upon failures, the extent is immediately sealed. This model allows the consistency to be enforced across all the replicas via their commit lengths.

Furthermore, the append-only system has allowed us to keep snapshots of the previous states at virtually no extra cost, which has made it easy to provide snapshot/versioning features. It also has allowed us to efficiently provide optimizations like erasure coding. In addition, append-only has been a tremendous benefit for diagnosing issues as well as repairing/recovering the system in case something goes wrong. Since the history of changes is preserved, tools can easily be built to diagnose issues and to repair or recover the system from a corrupted state back to a prior known consistent state. When operating a system at this scale, we cannot emphasize enough the benefit we have seen from using an append-only system for diagnostics and recovery.

An append-based system comes with certain costs. An efficient and scalable garbage collection (GC) system is crucial to keep the space overhead low, and GC comes at a cost of extra I/O. In addition, the data layout on disk may not be the same as the virtual address space of the data abstraction stored, which led us to implement prefetching logic for streaming large data sets back to the client.

End-to-end Checksums – We found it crucial to keep checksums for user data end to end. For example, during a blob upload, once the Front-End server receives the user data, it immediately computes the checksum and sends it along with the data to the backend servers. Then at each layer, the partition server and the stream servers verify the checksum before continuing to process it. If a mismatch is detected, the request is failed. This prevents corrupted data from being committed into the system. We have seen cases where a few servers had hardware issues, and our end-to-end checksum caught such issues and helped maintain data integrity. Furthermore, this end-to-end checksum mechanism also helps identify servers that consistently have hardware issues so we can take them out of rotation and mark them for repair.

Upgrades – A rack in a storage stamp is a fault domain. A concept orthogonal to fault domain is what we call an upgrade domain (a set of servers briefly taken offline at the same time during a rolling upgrade). Servers for each of the three layers are spread evenly across different fault and upgrade domains for the storage service. This way, if a fault domain goes down, we lose at most 1/X of the servers for a given layer, where X is the number of fault domains. Similarly, during a service upgrade at most 1/Y of the servers for a given layer are upgraded at a given time, where Y is the number of upgrade domains. To achieve this, we use rolling upgrades, which enable us to maintain high availability when upgrading the storage service, and we upgrade a single upgrade domain at a time. For example, if we have ten upgrade domains, then upgrading a single domain would potentially upgrade ten percent of the servers from each layer at a time.

During a service upgrade, storage nodes may go offline for a few minutes before coming back online. We need to maintain availability and ensure that enough replicas are available at any point in time. Even though the system is built to tolerate isolated failures, these planned (massive) upgrade "failures" can be more efficiently dealt with instead of being treated as abrupt massive failures. The upgrade process is automated so that it is tractable to manage a large number of these large-scale deployments. The automated upgrade process goes through each upgrade domain one at a time for a given storage stamp. Before taking down an upgrade domain, the upgrade process notifies the PM to move the partitions out of that upgrade domain and notifies the SM to not allocate new extents in that upgrade domain. Furthermore, before taking down any servers, the upgrade process checks with the SM to ensure that there are sufficient extent replicas available for each extent outside the given upgrade domain. After upgrading a given domain, a set of validation tests are run to make sure the system is healthy before proceeding to the next upgrade domain. This validation is crucial for catching issues during the upgrade process and stopping it early should an error occur.

Multiple Data Abstractions from a Single Stack – Our system supports three different data abstraction from the same storage stack: Blobs, Tables and Queues. This design enables all data abstractions to use the same intra-stamp and inter-stamp replication, use the same load balancing system, and realize the benefits from improvements in the stream and partition layers. In addition, because the performance needs of Blobs, Tables, and Queues are different, our single stack approach enables us to reduce costs by running all services on the same set of hardware. Blobs use the massive disk capacity, Tables use the I/O spindles from the many disks on a node (but do not require as much capacity as Blobs), and Queues mainly run in memory. Therefore, we are not only blending different customer's workloads together on shared resources, we are also blending together Blob, Table, and Queue traffic across the same set of storage nodes.

Use of System-defined Object Tables – We chose to use a fixed number of system defined Object Tables to build Blob, Table, and Queue abstractions instead of exposing the raw Object Table semantics to end users. This decision reduces management by our system to only the small set of schemas of our internal, system defined Object Tables. It also provides for easy maintenance and upgrade of the internal data structures and isolates changes of these system defined tables from end user data abstractions.

Offering Storage in Buckets of 100TBs – We currently limit the amount of storage for an account to be no more than 100TB. This constraint allows all of the storage account data to fit within a given storage stamp, especially since our initial storage stamps held only two petabytes of raw data (the new ones hold 20-30PB). To obtain more storage capacity within a single data center, customers use more than one account within that location. This

ended up being a reasonable tradeoff for many of our large customers (storing petabytes of data), since they are typically already using multiple accounts to partition their storage across different regions and locations (for local access to data for their customers). Therefore, partitioning their data across accounts within a given location to add more storage often fits into their existing partitioning design. Even so, it does require large services to have account level partitioning logic, which not all customers naturally have as part of their design. Therefore, we plan to increase the amount of storage that can be held within a given storage account in the future.

CAP Theorem – WAS provides high availability with strong consistency guarantees. This combination seems to violate the CAP theorem [2], which says a distributed system cannot have availability, consistency, and partition tolerance at the same time. However, our system, in practice, provides all three of these properties within a storage stamp. This situation is made possible through layering and designing our system around a specific fault model.

The stream layer has a simple append-only data model, which provides high availability in the face of network partitioning and other failures, whereas the partition layer, built upon the stream layer, provides strong consistency guarantees. This layering allows us to decouple the nodes responsible for providing strong consistency from the nodes storing the data with availability in the face of network partitioning. This decoupling and targeting a specific set of faults allows our system to provide high availability and strong consistency in face of various classes of failures we see in practice. For example, the type of network partitioning we have seen within a storage stamp are node failures and top-of-rack (TOR) switch failures. When a TOR switch fails, the given rack will stop being used for traffic — the stream layer will stop using that rack and start using extents on available racks to allow streams to continue writing. In addition, the partition layer will reassign its RangePartitions to partition servers on available racks to allow all of the data to continue to be served with high availability and strong consistency. Therefore, our system is designed to be able to provide strong consistency with high availability for the network partitioning issues that are likely to occur in our system (at the node level as well as TOR failures).

High-performance Debug Logging – We used an extensive debug logging infrastructure throughout the development of WAS. The system writes logs to the local disks of the storage nodes and provides a grep-like utility to do a distributed search across all storage node logs. We do not push these verbose logs off the storage nodes, given the volume of data being logged.

When bringing WAS to production, reducing logging for performance reasons was considered. The utility of verbose logging though made us wary of reducing the amount of logging in the system. Instead, the logging system was optimized to vastly increase its performance and reduce its disk space overhead by automatically tokenizing and compressing output, achieving a system that can log 100's of MB/s with little application performance impact per node. This feature allows retention of many days of verbose debug logs across a cluster. The high-performance logging system and associated log search tools are critical for investigating any problems in production in detail without the need to deploy special code or reproduce problems.

Pressure Point Testing – It is not practical to create tests for all combinations of all complex behaviors that can occur in a large scale distributed system. Therefore, we use what we call Pressure Points to aid in capturing these complex behaviors and interactions. The system provides a programmable interface for all of the main operations in our system as well as the points in the system to create faults. Some examples of these pressure point commands are: checkpoint a RangePartition, combine a set of RangePartition checkpoints, garbage collect a RangePartition, split/merge/load balance RangePartitions, erasure code or un-erasure code an extent, crash each type of server in a stamp, inject network latencies, inject disk latencies, etc.

The pressure point system is used to trigger all of these interactions during a stress run in specific orders or randomly. This system has been instrumental in finding and reproducing issues from complex interactions that might have taken years to naturally occur on their own.

9. Related Work

Prior studies [9] revealed the challenges in achieving strong consistency and high availability in a poorly-connected network environment. Some systems address this by reducing consistency guarantees to achieve high availability [22,14,6]. But this shifts the burden to the applications to deal with conflicting views of data. For instance, Amazon's SimpleDB was originally introduced with an eventual consistency model and more recently added strongly consistent operations [23]. Van Renesse et. al. [20] has shown, via Chain Replication, the feasibility of building large-scale storage systems providing both strong consistency and high availability, which was later extended to allow reading from any replica [21]. Given our customer needs for strong consistency, we set out to provide a system that can provide strong consistency with high availability along with partition tolerance for our fault model.

As in many other highly-available distributed storage systems [6,14,1,5], WAS also provides geo-redundancy. Some of these systems put geo-replication on the critical path of the live application requests, whereas we made a design trade-off to take a classical asynchronous geo-replication approach [18] and leave it off the critical path. Performing the geo-replication completely asynchronously allows us to provide better write latency for applications, and allows more optimizations, such as batching and compaction for geo-replication, and efficient use of cross-data center bandwidth. The tradeoff is that if there is a disaster and an abrupt failover needs to occur, then there is unavailability during the failover and a potential loss of recent updates to a customer's account.

The closest system to ours is GFS [8,15] combined with BigTable [4]. A few differences from these prior publications are: (1) GFS allows relaxed consistency across replicas and does not guarantee that all replicas are bitwise the same, whereas WAS provides that guarantee, (2) BigTable combines multiple tablets into a single commit log and writes them to two GFS files in parallel to avoid GFS hiccups, whereas we found we could work around both of these by using journaling in our stream layer, and (3) we provide a scalable Blob storage system and batch Table transactions integrated into a BigTable-like framework. In addition, we describe how WAS automatically load balances, splits, and merges RangePartitions according to application traffic demands.

10. Conclusions

The Windows Azure Storage platform implements essential services for developers of cloud based solutions. The combination of strong consistency, global partitioned namespace, and disaster recovery has been important customer features in WAS's multi-tenancy environment. WAS runs a disparate set of workloads with

various peak usage profiles from many customers on the same set of hardware. This significantly reduces storage cost since the amount of resources to be provisioned is significantly less than the sum of the peak resources required to run all of these workloads on dedicated hardware.

As our examples demonstrate, the three storage abstractions, Blobs, Tables, and Queues, provide mechanisms for storage and workflow control for a wide range of applications. Not mentioned, however, is the ease with which the WAS system can be utilized. For example, the initial version of the Facebook/Twitter search ingestion engine took one engineer only two months from the start of development to launching the service. This experience illustrates our service's ability to empower customers to easily develop and deploy their applications to the cloud.

Additional information on Windows Azure and Windows Azure Storage is available at http://www.microsoft.com/windowsazure/.

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